

Coarsening Confers Robustness: Causal Bayesian Optimization under DAG Misspecification

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Abstract

Causal Bayesian optimization (CBO) uses a causal graph to select intervention sets and construct priors over the interventional response. A wrong graph can therefore bias a prior or remove an optimal arm. We replace the full graph by a partition and its cluster-level quotient, and study behavior over the *protected class*: all fine graphs sharing that quotient. Our central result separates the regimes. A single quotient-invisible edge deletion forces any optimizer whose interventions and incumbent respect the wrong graph’s minimal intervention sets into arbitrarily large linear regret, while an optimizer that factors through the quotient has identical trajectories across the entire class—its worst-case regret over the class equals its regret under the true graph, decomposing exactly into the coarsening price plus within-quotient optimization regret. Conversely, any trajectory map invariant over every protected class factors through the quotient. Under full intervention support and cluster-contained confounding the price is zero; otherwise a positive action-space gap lower-bounds incumbent regret linearly in the horizon, and a heuristic refinement scheme can recover fine actions. We instantiate the contract for three families of the field: arm-based, dynamic, and model-based CBO. Experiments isolate prior and arm-set corruption, verify exact quotient invariance, and test all three families on their standard benchmarks.

1 Introduction

Causal Bayesian optimization [Aglietti et al., 2020] searches over interventions and uses observational data

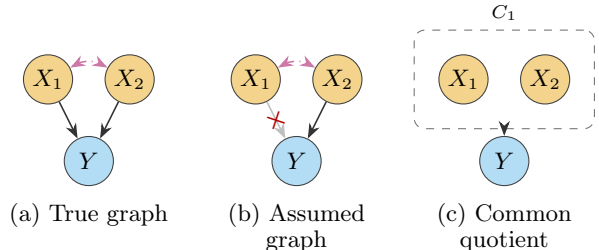


Figure 1: PARALLELPARENT. Deleting $X_1 \rightarrow Y$ changes the fine search space but not the quotient because $C_1 \rightarrow Y$ remains through $X_2 \rightarrow Y$. Orange nodes are manipulable, blue is the target, and the dash-dotted edge denotes latent confounding.

with a causal graph to build priors over the interventional response. CBO and its descendants [Aglietti et al., 2021, 2023, Sussex et al., 2023] usually assume that the full graph is correct. Yet changing only the assumed graph can substantially change their performance [Anonymous, 2026]. We study graph misspecification, where the learner trusts one graph that differs from the truth. Our premise is simple: commit only to structural distinctions that are reliable.

Wrong edges are costly: two-variable example. Let X_1 and X_2 be confounded parents of Y , with

$$Y = \lambda(X_1 - 2)^2 + (X_2 + 2)^2 + \varepsilon, \quad y^* = 0.$$

Deleting $X_1 \rightarrow Y$ from the assumed graph removes every X_1 -containing fine arm. The best remaining intervention satisfies

$$\min_{x_2} \mathbb{E}[Y \mid \text{do}(X_2 = x_2)] = 4.25\lambda.$$

For $C_1 = \{X_1, X_2\}$, however, both assumptions induce the same quotient $C_1 \rightarrow Y$ (Fig. 1). A quotient method therefore follows the same trajectory under both assumptions.

Quotient CBO. Full-DAG methods and online structure learning have both been studied [Mukherjee et al., 2024, Branchini et al., 2023]. We study

the middle ground. The practitioner supplies a partition Π and a cluster DAG (C-DAG) over its clusters. One cluster-level judgement replaces several edge-level judgements. QCBO applies CBO’s minimal-intervention-set rule to this quotient and uses identifiable cluster effects as prior means. Cluster-level identification is already well developed [Anand et al., 2023, Ferreira and Assaad, 2025, Wahl et al., 2024].

Contributions.

- **Separation over the protected class.** One quotient-invisible edge deletion costs fine-graph methods arbitrarily large linear worst-case regret, while a quotient method’s worst-case regret over the class equals its regret under the truth—an exact decomposition into coarsening price plus within-quotient regret (Theorem 1).
- **Input contract and characterization.** Trajectory invariance over the protected class holds for any optimizer that factors through the quotient; conversely, any trajectory map invariant over every protected class factors through the quotient (Lemma 1, Prop. 5).
- **Value and refinement.** We characterize the coarsening gap, give a graphical sufficient condition for zero gap (Def. 4, Cor. 1), show a positive gap lower-bounds incumbent regret linearly under a fixed partition (Prop. 6), and state conditional recovery by refinement (Cor. 2).
- **Experiments.** Minimal SCMs isolate the two misspecification channels and the coarsening gap. A portability suite evaluates QCBO, QDCBO, and QMCBO against their base methods.

2 Background

A structural causal model $\mathcal{M} = \langle \mathbf{U}, \mathbf{V}, F, P(\mathbf{U}) \rangle$ induces observational and interventional distributions [Pearl, 2009]. Its latent projection is an acyclic directed mixed graph (ADMG) $\mathcal{G} = (\mathbf{V}, E)$ with directed causal edges and bidirected latent-confounding edges [Lee and Bareinboim, 2019].

Causal Bayesian optimization. Given manipulable variables $\mathbf{X} \subseteq \mathbf{V}$ and target Y , CBO [Aglietti et al., 2020] solves

$$(\mathbf{X}_s^*, \mathbf{x}_s^*) = \arg \min_{\substack{\mathbf{X}_s \in \mathbf{ES} \\ \mathbf{x}_s \in D(\mathbf{X}_s)}} \mathbb{E}[Y \mid \text{do}(\mathbf{X}_s = \mathbf{x}_s)], \quad (1)$$

where $D(\mathbf{X}_s)$ is the intervention domain. CBO maintains one Gaussian process (GP) per candidate intervention set. Its exploration set contains the minimal intervention sets (MIS) of the assumed graph [Lee and

Bareinboim, 2018, Aglietti et al., 2020]. Writing $\mathcal{G}_{\bar{\mathbf{X}}_s}$ for the graph without incoming edges to $\mathbf{X}_s$ and $\text{An}(Y)$ for the ancestors of Y ,

$$\mathbf{ES} = \left\{ \mathbf{X}_s : \begin{array}{l} \emptyset \neq \mathbf{X}_s \subseteq \mathbf{X}, \\ \mathbf{X}_s \subseteq \text{An}(Y) \text{ in } \mathcal{G}_{\bar{\mathbf{X}}_s} \end{array} \right\}. \quad (2)$$

an arm is redundant when one of its variables ceases to be an ancestor of Y after intervention. Each GP uses a do-calculus estimate as prior mean, when available. Thus a structural hypothesis enters through

$$\mathcal{G} \mapsto (\mathbf{ES}(\mathcal{G}), \{m_s^{\mathcal{G}}\}_{s \in \mathbf{ES}(\mathcal{G})}) \mapsto \tau,$$

where τ is the optimization trajectory. A wrong graph can change a prior or the arm set itself.

Coarsening and C-DAGs.

Definition 1 (Coarsening [Madaleno et al., 2026a]). *For a partition Π of $\mathbf{V}$, let $\chi : \mathbf{V} \rightarrow \Pi$ map each vertex to its cluster. Its directed quotient has*

$$E^\Pi = \{(\chi(u), \chi(v)) : (u, v) \in E, \chi(u) \neq \chi(v)\}.$$

Bidirected edges are mapped in the same way. The coarsening is valid when the directed quotient is acyclic.

$$Q_\Pi(\mathcal{G}) := (\Pi, \mathcal{G}^\Pi).$$

Define the coarse action family by

$$\mathcal{U}(\Pi) = \left\{ \bigcup_{C \in S} C : \emptyset \neq S \subseteq \Pi_M \right\},$$

where Π_M contains the manipulable clusters. For $A \subseteq \mathbf{X}$, define

$$\mu^*(A) := \min_{a \in D(A)} \mathbb{E}[Y \mid \text{do}(A = a)],$$

$$V(\Pi) := \min_{A \in \mathcal{U}(\Pi)} \mu^*(A),$$

$$\Delta(\Pi) := V(\Pi) - V(\Pi_\perp).$$

and let $\Pi_\perp$ be the singleton partition. The MIS reduction in Lemma 2 connects this value to QCBO:

$$\min_{A \in \mathbf{ES}^\Pi} \mu^*(A) = \min_{A \in \mathcal{U}(\Pi)} \mu^*(A) = V(\Pi). \quad (3)$$

C-DAG identification is sound and complete over the fine graphs represented by the quotient [Anand et al., 2023]. Cluster-level Markov properties transfer, while faithfulness need not [Wahl et al., 2024].

Evaluation metrics. Let $y^* := V(\Pi_\perp)$ and let $\hat{a}_t$ be the incumbent after trial t . Incumbent mean regret is

$$r_t := \mu(\hat{a}_t) - y^* \geq 0, \quad R_T := \sum_{t=1}^T r_t.$$

The experiments also report paired trajectory differences between correct and misspecified inputs.

3 What coarsening buys

Write $\Pi' \preceq \Pi$ when every cluster of Π' is contained in a cluster of Π . An optimizer satisfies the quotient contract when its graph-dependent components use $\mathcal{G}$ only through $Q_\Pi(\mathcal{G})$.

Lemma 1 (Quotient contract). *If*

$$\mathcal{A}(\mathcal{G}, \mathcal{D}, \omega, \eta) = \bar{\mathcal{A}}(Q_\Pi(\mathcal{G}), \mathcal{D}, \omega, \eta),$$

then, for fixed data $\mathcal{D}$, random stream ω , and hyperparameters η ,

$$Q_\Pi(\mathcal{G}) = Q_\Pi(\mathcal{G}') \implies \tau_{\mathcal{A}}(\mathcal{G}) = \tau_{\mathcal{A}}(\mathcal{G}').$$

This invariance is immediate—by design—so we state it as a lemma: the *contract* our constructions must honor, not a result in itself. The substance lies in what honoring it costs (§3.1) and what it buys (§3.2). The object it protects against is a class of fine graphs:

Definition 2 (Protected class). *For an ADMG $\mathcal{G}$ over $\mathbf{V}$ and a valid partition Π ,*

$$E_\Pi(\mathcal{G}) := \{\mathcal{G}' \text{ ADMG over } \mathbf{V} : Q_\Pi(\mathcal{G}') = Q_\Pi(\mathcal{G})\}.$$

Within-cluster edits and quotient-redundant cross-cluster edits stay inside $E_\Pi(\mathcal{G})$ and are protected. Quotient-visible edits leave the class and are not.

3.1 What does protection cost?

Proposition 1 (Validity is not refinement-monotone).

$$\exists \Pi' \preceq \Pi : \mathcal{G}^\Pi \text{ acyclic}, \quad \mathcal{G}^{\Pi'} \text{ cyclic}.$$

Validity must therefore be checked after refinement.

Proposition 2 (Refinement preserves retained arms). *Let $\Pi' \preceq \Pi$ be valid. If $A \in \mathbf{ES}^\Pi \cap \text{MIS}(\mathcal{G}^{\Pi'})$, then*

$$A \in \mathbf{ES}^{\Pi'}, \quad \phi_A^{\Pi'}(P) = P(Y \mid \text{do}(A)).$$

Thus the coarse identification functional and the causal value remain valid under $\mathcal{G}^{\Pi'}$. MIS membership itself need not be monotone.

Proposition 3 (Monotonicity and price of coarsening). *For valid $\Pi' \preceq \Pi$,*

$$\mathcal{U}(\Pi) \subseteq \mathcal{U}(\Pi'), \quad 0 \leq \Delta(\Pi') \leq \Delta(\Pi).$$

Moreover,

$$\Delta(\Pi) = 0 \iff \arg \min_{A \in \mathcal{U}(\Pi_\perp)} \mu^*(A) \cap \mathcal{U}(\Pi) \neq \emptyset.$$

Proposition 4 (Sufficient conditions for zero coarsening gap). *Assume a semi-Markovian SCM. For every $A \subseteq \mathbf{X}$, define*

$$\begin{aligned} \text{cl}_\Pi(A) &:= \bigcup \{C \in \Pi : C \cap A \neq \emptyset\}, \\ B &:= \text{cl}_\Pi(A) \setminus A. \end{aligned}$$

For each $a \in D(A)$ let B_a be the joint natural response under $\text{do}(A = a)$ and set

$$D_B(a) := \{b : (a, b) \in D(\text{cl}_\Pi(A))\}.$$

If, for all A, a , the support inclusion and cross-world independence premises

$$\text{supp}(B_a) \subseteq D_B(a), \quad B_a \perp Y_{a,b} \quad \forall b \in D_B(a)$$

hold, then $V(\Pi) = V(\Pi_\perp)$ and $\Delta(\Pi) = 0$.

Both premises deserve names at the level a practitioner can check. The first is a domain condition:

Definition 3 (Full intervention support). *Intervention domains are products, $D(W) = \prod_{w \in W} D(w)$ for every $W \subseteq \mathbf{X}$, and for every manipulable v and every intervention a on any $A \subseteq \mathbf{X} \setminus \{v\}$, $\text{supp}(v_a) \subseteq D(v)$: each variable’s intervention domain contains every value it attains as a natural response.*

The second premise couples potential outcomes under different interventions. We give a graph-level condition that implies it, stated where the proof actually operates: in the augmented NPSEM graph.

Definition 4 (Cluster-contained confounding). *Represent the semi-Markovian SCM by an NPSEM with mutually independent exogenous variables, one shared exogenous variable per bidirected edge, and augmented graph $\mathcal{G}^{\text{aug}}$. For $A \subseteq \mathbf{X}$ with $B = \text{cl}_\Pi(A) \setminus A$, let $\mathbf{U}_B(A)$ be the exogenous ancestors of B in $\mathcal{G}^{\text{aug}}$ mutilated by $\text{do}(A)$ (the members of B respond naturally and are not clamped), and let $\mathbf{U}_Y(A)$ be the exogenous ancestors of Y in $\mathcal{G}^{\text{aug}}$ mutilated by $\text{do}(\text{cl}_\Pi(A))$. The partition Π has cluster-contained confounding when $\mathbf{U}_B(A) \cap \mathbf{U}_Y(A) = \emptyset$ for every $A \subseteq \mathbf{X}$.*

Informally: once a cluster union is clamped, no latent influence is shared between the un-clamped remainder of a cluster and the target. The familiar ADMG reading—every backdoor path from a cluster member to Y meets another member of that cluster before reaching Y —is sufficient for Def. 4 (App. A); the exogenous-ancestor statement is the condition the proof consumes.

Corollary 1 (Graphical zero-gap condition). *In a semi-Markovian model, full intervention support (Def. 3) implies support inclusion, and cluster-contained confounding (Def. 4) implies cross-world independence (Lemma 3). Hence the premises of Prop. 4 hold and $\Delta(\Pi) = 0$.*

Remark 1 (On the cross-world premise). $B_a \perp Y_{a,b}$ is untestable from any single-world data, observational or from any one experiment. Cor. 1 therefore never invokes it as a primitive: within an NPSEM representation with mutually independent exogenous variables it is derived, via Lemma 3, from Def. 4, an ordinary structural judgement about where confounding

lives. The premise inherits exactly the credibility of that judgement, no more.

Within this model class, the contrapositive is

$$\Delta(\Pi) > 0 \implies \begin{array}{c} \text{support failure or} \\ \text{confounding escaping a cluster.} \end{array}$$

Thus coarsening need not sacrifice the best attainable value. Even when $\Delta(\Pi) = 0$, it may change finite-sample regret, dimension, and intervention cost.

3.2 What protection buys: worst-case regret

Fix a true SCM $\mathcal{M}$ with latent projection $\mathcal{G}$, and hand the optimizer a *hypothesis* graph H while every intervention outcome is drawn from $\mathcal{M}$ (the fixed-objective seam of §5; App. C). Write $R_T(\mathcal{A}; H)$ for the incumbent regret of §2 when $\mathcal{A}$ runs with input H at fixed $(\mathcal{D}, \omega, \eta)$, and

$$\text{WR}_T(\mathcal{A}) := \sup_{H \in E_\Pi(\mathcal{G})} R_T(\mathcal{A}; H)$$

for the worst case over the protected class.

Theorem 1 (Protected-class regret separation). (i)

Fine-graph fragility. For every $\beta > 0$ there exist a semi-Markovian SCM $\mathcal{M}_\lambda$ (PARALLEL-PARENT with $\lambda = \beta/4.25$, App. B), a valid Π with $\Delta(\Pi) = 0$, and $H \in E_\Pi(\mathcal{G})$ obtained from $\mathcal{G}$ by deleting a single directed edge, such that every optimizer whose executed intervention sets—initial design and sequential—all lie in $\text{MIS}(H)$, and whose incumbent is selected among those executed interventions, satisfies $r_t \geq \beta$ for all t , hence $\text{WR}_T \geq \beta T$.

(ii) **Uniform protection with exact decomposition.** For every $\mathcal{M}$, valid Π , and fixed $(\mathcal{D}, \omega, \eta)$: if $\mathcal{A}$ satisfies the quotient contract for Π , executes only interventions on cluster unions $A \in \mathcal{U}(\Pi)$, and selects its incumbent among executed actions, then its trajectory is identical for every $H \in E_\Pi(\mathcal{G})$ and

$$\begin{aligned} \text{WR}_T(\mathcal{A}) &= R_T(\mathcal{A}; \mathcal{G}) \\ &= T \Delta(\Pi) + \sum_{t=1}^T (\mu(\hat{a}_t) - V(\Pi)), \end{aligned}$$

with both terms nonnegative: an unavoidable per-trial structural gap, plus within-quotient optimization regret that no member of the protected class can alter.

Part (i) is the headline: a single quotient-invisible edge deletion costs a fine-graph method an arbitrarily large linear regret while the quotient loses nothing ($\Delta(\Pi) = 0$). Experiment A instantiates the construction at $\lambda = 1$ (§5.1). Part (ii) prices the protection exactly. A converse closes the design space:

Proposition 5 (The contract is necessary for trajectory-level protection). If the trajectory map $H \mapsto \tau(H)$ at fixed $(\mathcal{D}, \omega, \eta)$ is constant on $E_\Pi(\mathcal{G})$ for every ADMG $\mathcal{G}$, then it factors through Q_Π .

This characterizes the *trajectory map*, not an optimizer’s internal implementation: any procedure whose behavior is invariant over every protected class is, at the trajectory level, a function of the quotient alone.

Refining the partition: Hierarchical QCBO (HQCBO).

Proposition 6 (Linear regret floor under a fixed partition). For a fixed partition,

$$r_t \geq \Delta(\Pi) \text{ for every } t, \quad R_T \geq T \Delta(\Pi).$$

The floor concerns the incumbent mean, not the exploratory action at trial t : a positive action-space gap contributes linearly in the horizon, and no within-quotient learning removes it. Refinement can. HQCBO starts at Π and splits the incumbent’s clusters along a declared hierarchy when progress stalls; it rejects refinements whose quotient is cyclic (Prop. 1), retains old-arm data, and initializes only new arms.

Definition 5 (Incumbent consistency). The base optimizer’s incumbent is consistent on the action space of a partition Π_J if, when the optimizer continues on that action space, $\mu(\hat{a}_t) \rightarrow V(\Pi_J)$ almost surely as $t \rightarrow \infty$.

Corollary 2 (Conditional refinement recovery). If the refinement schedule visits a valid Π_J with $\Delta(\Pi_J) = 0$ and the incumbent is consistent on Π_J ’s action space (Def. 5), then $r_t \rightarrow 0$ and $R_T/T \rightarrow 0$.

Whether Π_J is visited depends on the trigger, a *heuristic* with no guarantee: HQCBO splits when the best-so-far value has improved by less than δ over the last k trials (exact rule and defaults in App. C). Cor. 2 is conditional on the trigger firing, on the declared hierarchy exposing an optimal arm, and on consistency—a property of the base optimizer that we assume, not prove; with continuous within-arm domains and unbounded noise it fails for, e.g., raw running minima of noisy observations.

Corollary 3 (Adaptive refinement invariance). Write $\tau_{\mathcal{H}}$ for the HQCBO trajectory under a declared hierarchy $\mathcal{H}$. For visited partitions $\Pi_0, \dots, \Pi_J$,

$$[\forall j, \mathcal{G}^{\Pi_j} = (\mathcal{G}')^{\Pi_j}] \implies \tau_{\mathcal{H}}(\mathcal{G}) = \tau_{\mathcal{H}}(\mathcal{G}').$$

Splitting a cluster exposes its internal edges in later phases. Edits inside clusters that are never split remain invisible.

4 Instantiations: QCBO, QDCBO, and QMCBO

Each construction replaces the base method’s graph-coupled components by their quotient counterparts. Appendix C gives estimator and implementation details.

4.1 Arm-based: quotient CBO (QCBO)

QCBO receives a partition of the manipulable variables and its C-DAG. The target and non-manipulable variables remain singletons. Define $U_\Pi(S) := \bigcup_{C \in S} C$. Applying Eq. (2) to the quotient gives

$$\mathbf{ES}^\Pi = \left\{ U_\Pi(S) : \begin{array}{l} \emptyset \neq S \subseteq \Pi_M, \\ S \subseteq \text{An}(Y) \text{ in } \mathcal{G}_S^\Pi \end{array} \right\}. \quad (4)$$

Each arm sets one or more clusters jointly.

Identifiability and the two-tier prior. For each $A \in \mathbf{ES}^\Pi$, the complete ID algorithm [Shpitser and Pearl, 2006, Ananke] determines the prior:

$$m_A(a) = \begin{cases} \hat{\mathbb{E}}[Y \mid \text{do}(A = a)] & \text{if identified in } \mathcal{G}^\Pi, \\ \bar{Y}_{\text{obs}} & \text{otherwise.} \end{cases} \quad (5)$$

Non-identifiable arms learn from experimental data. Identification from the C-DAG is sound for every represented fine graph [Anand et al., 2023].

Assumption 1 (Estimator boundary). *The quotient is valid. The ID test is complete, while our numerical estimator is sound but not complete. Identifiable expressions outside its implemented class receive the uninformative prior.*

Proposition 7 (Identity recovery).

$$\begin{aligned} \Pi = \Pi_\perp &\implies \mathcal{G}^\Pi = \mathcal{G}^*, \\ \mathbf{ES}^\Pi &= \text{MIS}(\mathcal{G}^*), \quad m_A^\Pi = m_A^{CBO}. \end{aligned}$$

Identity is with respect to CBO *as specified*—the MIS exploration set with do-calculus priors. The authors’ released implementation deviates from that specification in two documented places, which we correct for both methods (App. C).

4.2 Dynamic: quotient DCBO (QDCBO)

Dynamic CBO (DCBO) optimizes interventions across a dynamic Bayesian network [Aglietti et al., 2021]. QDCBO quotients each slice and its cross-slice transitions, then replaces emission and transition models by joint cluster mechanisms. Its exploration sets use Eq. (4) per slice. Information propagates to the next slice exactly as in DCBO.

Corollary 4 (QDCBO inherits the contract). *QDCBO satisfies Lemma 1 per slice. At $\Pi = \Pi_\perp$ it equals DCBO, and Prop. 3 applies to each slice’s reachable interventions.*

4.3 Model-based: quotient MCBO (QMCBO)

Model-based CBO (MCBO) composes one GP per causal mechanism and selects interventions using the Gaussian-process upper confidence bound acquisition function (GP-UCB) [Srinivas et al., 2010, Sussex et al., 2023]. QMCBO replaces the node mechanisms by

$$P(\mathbf{X}_C \mid \mathbf{X}_{\text{pa}(C)}),$$

modeled as multi-output GPs with a coregionalization kernel [Bonilla et al., 2007]. These joint mechanisms are composed along the C-DAG.

Corollary 5 (QMCBO inherits the contract). *QMCBO satisfies Lemma 1. At $\Pi = \Pi_\perp$ it equals MCBO. Its reachable interventions are whole-cluster patterns, so Prop. 3 also applies.*

Proposition 8 (Soundness of cluster mechanisms). *Under causal sufficiency and valid Π , for every union of clusters S ,*

$$P(\mathbf{V} \mid \text{do}(\mathbf{x}_S)) = \prod_{C \notin S} P(\mathbf{X}_C \mid \mathbf{X}_{\text{pa}(C)}) \Big|_{\mathbf{x}_S = \mathbf{x}_S}.$$

The mechanism must be joint to retain within-cluster dependence.

Proposition 9 (Exposure: sub-cluster interventions are not identifiable from cluster mechanisms). *There exist causally sufficient SCMs M_1, M_2 , valid for the same Π , such that*

$$\begin{aligned} \mathcal{I}_\Pi(M_1) &= \mathcal{I}_\Pi(M_2), \\ P^{M_1}(Y \mid \text{do}(x_A)) &\neq P^{M_2}(Y \mid \text{do}(x_A)) \end{aligned}$$

for a strict subset $A \subsetneq C$. Here $\mathcal{I}_\Pi$ contains the joint cluster mechanisms and all whole-cluster interventional distributions.

Supplying the input in practice. The contract’s input is a partition of the variables plus a C-DAG over its clusters—never a fine graph. Clusters typically follow domain modules (subsystems, assays, market segments, pathways); each C-DAG edge is then a single cluster-level judgement (“can subsystem A affect B ?”), one per pair, in place of many edge-level judgements. Validity is a syntactic acyclicity check (Def. 1), re-run after every refinement (Prop. 1). In restricted model classes the consistency of a proposed coarsening is itself testable [Madaleno et al., 2026b]. A refinement hierarchy, when supplied, follows the same nested domain taxonomy; HQCBO never needs distinctions finer than the hierarchy declares.

5 Experiments

We first isolate quotient invariance, prior corruption, and the action-space gap on three SCMs with closed-form oracles. We then test all three optimizer families on their standard benchmarks. Full specifications are in App. B.

Protocol. For each minimal SCM we fix the true mechanisms, observational data ($n_{\text{obs}} = 100$), seeds, and evaluation budget. Only the assumed structure changes. We use 30 paired seeds, $T = 50$ trials (60 for MEDIATEDCHAIN), three initial interventional points per arm, and per-variable unit intervention costs: an arm’s per-trial cost is its number of intervened variables, and acquisition divides expected improvement by this cost (Alg. 1). Initialization is unlogged and its budget differs across methods with different arm counts (e.g. 12 vs. 6 cost units on PARALLEL-PARENT); reported cumulative costs exclude it, and App. E quantifies the totals and re-compares methods at matched initialization-inclusive budgets. HQCBO uses the plateau trigger of App. C with $k = 5$ and $\delta = 10^{-3}$. The empirical cumulative-regret curves use the recorded best-so-far trajectory against the analytic y^* . They are descriptive and are not direct estimates of the mean-regret bound. Invariance is measured by $\sup_{s,t} |\Delta_{s,t}|$ between paired trajectories.

5.1 Experiment A: exploration-set corruption

PARALLEL-PARENT uses the SCM from Fig. 1 with $\lambda = 1$. The fine MIS is $\{\{X_1\}, \{X_2\}, \{X_1, X_2\}\}$ and $y^* = 0$. Deleting $X_1 \rightarrow Y$ leaves only $\{X_2\}$, whose optimum is 4.25. Fine CBO therefore plateaus at the analytic gap (Fig. 2a): Experiment A instantiates Theorem 1(i) at $\lambda = 1$. For $\Pi = \{\{X_1, X_2\}\}$ the deletion is quotient-invisible, and every QCBO trajectory is identical under the two graphs ($\sup |\Delta| = 0$). A spurious $X_1 \leftrightarrow Y$ is a quotient-visible negative control and changes the QCBO trajectory.

5.2 Experiment B: prior corruption without arm loss

FRONTDOOR uses $X_1 \rightarrow M \rightarrow Y$, latent $X_1 \leftrightarrow Y$, and a narrow-well objective whose optimum is $\text{do}(M = 1)$. Adding a spurious $X_1 \leftrightarrow M$ preserves the fine MIS but removes its identified priors. Fine CBO pays an empirical $\Delta R_{50} = 6.1 \pm 1.0$ and then reaches a similar final value (0.006 versus 0.001). Under $\Pi = \{\{X_1, M\}\}$ the edit is internal. QCBO is identical across conditions ($\sup |\Delta| = 0$).

5.3 Experiment C: the price of coarsening, and buying it back

MEDIATEDCHAIN has $X_1 \rightarrow X_2 \rightarrow Y$, $X_2 = 2X_1 + \varepsilon_2$, $Y = (X_2 - 4)^2$, and $D(X_2) = [-2, 2]$. For $\Pi = \{\{X_1, X_2\}\}$,

$$V(\Pi) = 4, \quad y^* = 0.04, \quad \Delta(\Pi) = 3.96.$$

The support condition in Prop. 4 fails because $\text{do}(X_1 = 2)$ lets X_2 respond near 4, outside its intervention box. CBO, QCBO, and HQCBO finish at 0.040, 4.000, and 0.040. HQCBO refines at trial 7.5 ± 0.1 . The fixed-partition empirical regret tracks the floor of Prop. 6, while the refined curve flattens after the fine arm appears: $R_{60} = 5.60 \pm 1.69$ (CBO), 243.23 ± 0.91 (QCBO, against the floor $T\Delta(\Pi) = 237.6$), and 37.22 ± 1.21 (HQCBO).

5.4 Experiment D: portability across families

We compare each quotient method with its base in the base paper’s released stack and metric. The suite contains three CBO tasks (10 seeds), three DCBO tasks (20 seeds), and two MCBO tasks (20 seeds). Graphs and full protocols are in Apps. D and E.

Results. Table 2 reports finals and per-task *paired* effects with 95% confidence intervals. The quotient improves the final value on stat (-0.29 , CI $[-0.41, -0.17]$), ind (-2.45 , CI $[-2.72, -2.19]$), and MCBO ToyGraph (-0.77 , CI $[-1.04, -0.50]$; $n=20$), where QMCBO reaches 2.161 ± 0.004 against MCBO’s 1.392 ± 0.128 . It pays a clear price on Synthetic ($+1.89$, CI $[+1.40, +2.39]$), whose optimum splits a cluster, and on Coral ($+0.34$, CI $[+0.30, +0.37]$). It is statistically indistinguishable from its base on Toy ($+0.03$, CI $[-0.04, +0.11]$) and nonstat (-1.69 , CI $[-3.57, +0.18]$); on PSAGraph the final difference is negligible at displayed precision while trajectories differ, and QMCBO uses $2.8\times$ less wall-clock time. At matched initialization-inclusive budgets the picture shifts further (App. E): the Synthetic price disappears, and Coral’s base has not completed initialization within the quotient’s entire budget. Under quotient-invisible perturbations QMCBO’s trajectories are identical on all 10 environment-seed pairs, while MCBO’s differ on 31 of 40; all twenty perturbed PSAGraph MCBO runs differ mid-run yet end within 0.001 of the same final (both saturate at -5.15), so Table 6 reports trajectory identity and final deviation separately. QDCBO is likewise invariant to its intra-slice edit.

Graph-uncertain comparator (CEO). We also run CEO [Branchini et al., 2023] on the minimal suite (Table 7, App. F) with a candidate pool holding the as-

Table 1: By Lemma 1, rows with an unchanged quotient are predicted invariant. The final column reports the measured paired deviation. PP = PARALLEL PARENT, FD = FRONT DOOR.

Cond.	Edit (assumed graph)	Quotient changed	Fine MIS changed	Fine prior changed	sup $ \Delta $
A1 (PP)	del $X_1 \rightarrow Y$ (quotient-redundant)	$\times$	$\checkmark$	—	0
A2 (PP)	add $X_1 \rightarrow X_2$ (intra-cluster bow)	$\times$	$\times$	$\checkmark$	0
A3 (PP)	assume $X_1 \leftrightarrow Y$ (quotient-visible)	$\checkmark$	$\times$	$\checkmark$	21.4
B1 (FD)	assume $X_1 \leftrightarrow M$ (intra-cluster conf.)	$\times$	$\times$	$\checkmark$	0

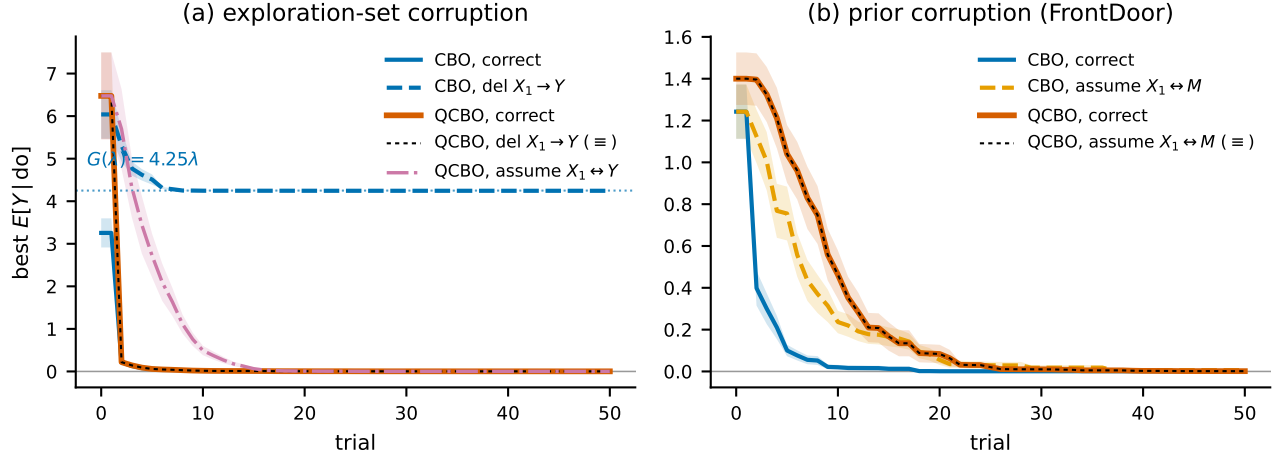


Figure 2: Correct and misspecified trajectories, mean $\pm$ s.e. over 30 seeds. (a) A wrong fine graph removes the optimal arm, while QCBO is unchanged for the quotient-invisible edit. (b) Prior corruption slows fine CBO transiently, while QCBO remains exactly invariant.

sorted observable DAG plus every observable-distinct variant. Shared with the fine methods: seeds, trial counts, three initial points per arm, the observational data, and the objective surface. Not shared: total initialization cost (CEO initializes every subset—12 units on PARALLEL PARENT against QCBO’s 6), a different acquisition objective (with the same per-variable cost normalization), and learning from single stochastic draws of the true SCM (its native regime) while incumbents are scored on the same mean surface; App. E reports initialization-inclusive totals and matched-budget values. Online structure learning does buy robustness: under the A1 deletion, hedged CEO ends at 0.77 where committed fine CBO plateaus at 4.245. But it pays a clear premium over the quotient on every protected condition (+0.07 to +1.00 at $n=30$, while QCBO is exactly invariant), its posterior mass on the baseline observable DAG (the truth’s directed edges with the bidirected edge dropped; no candidate can represent the latent confounding) plateaus near 0.54, and it cannot represent the assumed-confounder errors A3/B1 at all. On the unprotected price condition the picture mirrors: CEO’s fine action space reaches 0.135 where fixed-

partition QCBO pays the full gap (4.000), though HQCBO recovers further (0.040). Committing to a quotient and learning over fine graphs are complementary: each wins exactly where its structural assumption binds.

6 Related work

Graph uncertainty in CBO. QCBO neither assumes a full graph nor learns one. GACBO maintains a posterior over DAGs and searches the union of their MIS [Mukherjee et al., 2024]; our cluster-union MIS is the one-shot, coarse-input analogue of that union. CEO reduces structural uncertainty online through graph entropy [Branchini et al., 2023], and Durand et al. [2025] identify $\text{Pa}(Y)$ under a Y -leaf assumption; both are complementary rather than competing—an online learner can operate at cluster granularity, and its posterior could refine the quotient our contract consumes (measured under a shared protocol in App. F). Park et al. assume a Markov equivalence class [Park et al.]; the protected class $E_{\Pi}(\mathcal{G})$ is the analogous object for coarsening, with membership decided by the quotient rather than by conditional in-

Table 2: Family suite: finals and per-task *paired* final-performance effects Δ (quotient – base in the task’s own direction; positive = quotient worse) with paired- t 95% CIs. These are finite-budget empirical effects, not estimates of the action-space gap $\Delta(\Pi)$.

Experiment	Base final	Quotient final	Paired effect Δ [95% CI]
<i>CBO vs QCBO (min; $n=10$ paired seeds)</i>			
Toy	-2.16 ± 0.00	-2.13 ± 0.03	$+0.03 [-0.04, +0.11]$
Synthetic	-3.14 ± 0.19	-1.25 ± 0.07	$+1.89 [+1.40, +2.39]$
Coral	36.08 ± 0.02	36.42 ± 0.01	$+0.34 [+0.30, +0.37]$
<i>DCBO vs QDCBO (min; $n=20$ paired seeds)</i>			
stat ($T=3$)	-6.14 ± 0.05	-6.43 ± 0.02	$-0.29 [-0.41, -0.17]$
ind ($T=3$)	-3.12 ± 0.06	-5.57 ± 0.15	$-2.45 [-2.72, -2.19]$
nonstat ($T=3$)	8.03 ± 0.90	6.33 ± 0.29	$-1.69 [-3.57, +0.18]$
<i>MCBO vs QMCBO (max; $n=20$ paired seeds)</i>			
ToyGraph	1.39 ± 0.13	2.16 ± 0.00	$-0.77 [-1.04, -0.50]$
PSAGraph	-5.15 ± 0.00	-5.15 ± 0.00	$+0.0001 [-0.0004, +0.0006]$

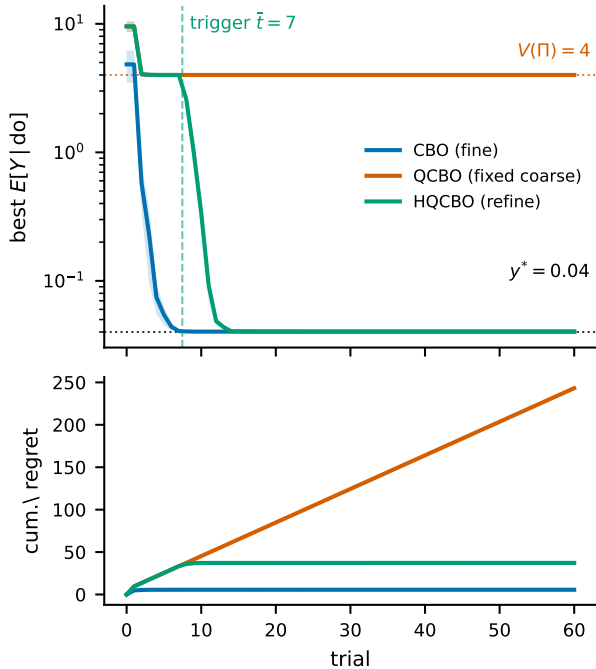


Figure 3: MEDIATEDCHAIN over 30 seeds. Top: mean $\pm$ s.e. best-so-far value and mean refinement time. Bottom: mean empirical cumulative regret, excluding the shared initial incumbent. Refinement exposes the fine optimal arm.

dependencies. Table 3 maps the nearest methods.

Causal bandits under structural uncertainty. Causal bandits bound regret with unknown or partial graphs: without the graph [Lu et al., 2021], bypassing graph learning [Konobeev et al., 2025], or with unobserved confounders under additive assumptions [Maiti et al., 2022]. Those guarantees price *learning* structure inside the horizon; Theorem 1 instead prices *commit-*

ting to a fixed coarse input, worst-case over the protected class, with no learning assumption.

Cluster-level identification. Anand et al. define C-DAGs and give sound and complete identification over them [Anand et al., 2023]; our two-tier prior consumes exactly that layer—the contribution here is the optimization-level contract and separation built on top, not the identification theory. Identifiability under partial observability and surrogate experiments [Lee and Bareinboim, 2020, Lee et al., 2020] governs which cluster queries a sound estimator may instantiate (Assumption 1). Macro-level causal structure can itself be learned [Ferreira and Assaad, 2025], though cluster-level faithfulness need not transfer [Wahl et al., 2024].

Abstraction and coarsening. Def. 1 instantiates constructive causal abstraction [Beckers and Halpern, 2019]: protection is invariance at the abstract level. Coarsening theory gives conditions under which a proposed clustering is consistent, including testable ones in restricted model classes [Madaleno et al., 2026a,b]. MFACBO coarsens the *outcome* into a low-fidelity proxy [Zeitler, 2025]; we coarsen manipulable inputs.

7 Discussion and limitations

The guarantee is behavioral and requires a correct quotient: arbitrary quotient-visible errors remain harmful, and nothing here protects against a wrong partition or a wrong C-DAG. The zero-gap condition rests on premises about the *true* model whose cross-world consequence no single-world data can test (Remark 1). Faithfulness need not transfer to clusters [Wahl et al., 2024]. The refinement trigger is a heuristic with no firing guarantee, so Cor. 2 is conditional on it. The experiments use per-variable unit intervention costs—cluster arms pay proportionally more per trial—and

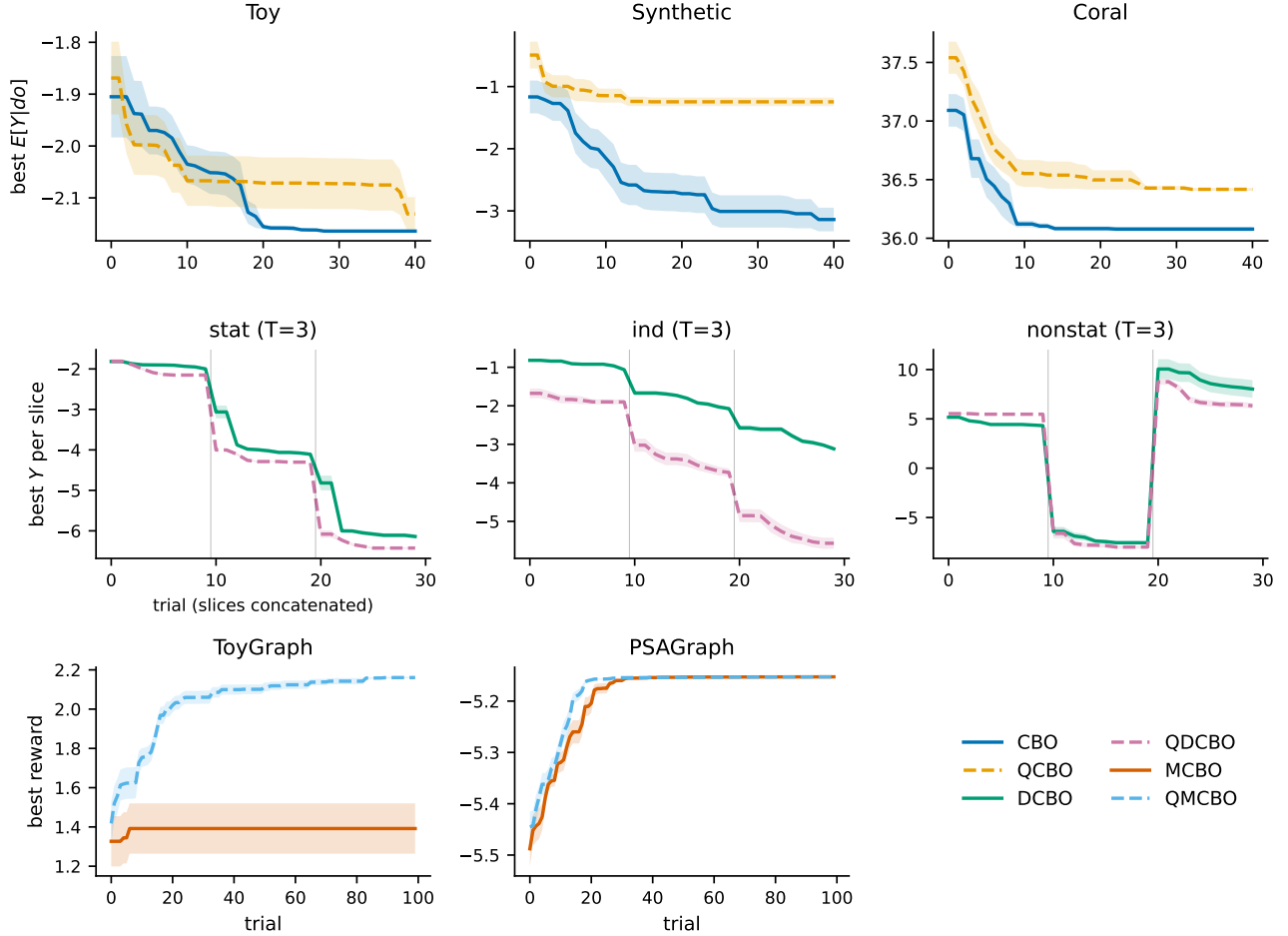


Figure 4: Quotient methods and their bases on the base papers’ benchmarks, mean $\pm$ s.e. Top: minimization for CBO. Middle: DCBO slices. Bottom: maximization for MCBO.

Table 3: Positioning of QCBO against the nearest methods.

Method	What is learned	Structural prior	Search space
CBO [Aglietti et al., 2020]	nothing (oracle DAG)	full DAG	MIS
GACBO [Mukherjee et al., 2024]	posterior over DAGs	DAG family	union of MIS
Durand et al. [Durand et al., 2025]	$\text{Pa}(Y) \subseteq \mathbf{X}$	Y leaf	$\mathbf{X}$
Park et al. [Park et al.]	$\text{PAG} \rightarrow \text{MAG}$	MEC	partial $\mathbf{X}$
SCB [Lee and Bareinboim, 2018]	nothing (oracle DAG)	full DAG	POMIS
QCBO (ours)	nothing	partition + C-DAG	MIS of the C-DAG (whole-cluster unions)

initialization budgets differ across methods with different arm counts; App. E quantifies both and re-compares at matched total budgets. A CEO comparison under the shared minimal-suite protocol is reported in §5.4 and App. F; GACBO, random-graph misspecification sweeps, and learned partitions remain deferred. Constructing the exploration set requires at most $2^K - 1$ MIS and ID checks for K manipulable clusters. QMCBO, like MCBO, assumes causal sufficiency. A wrong prior can be overcome if its arm remains available, while a removed or unavail-

able arm cannot be recovered without changing the action space. Learning the hierarchy and controlling unnecessary-refinement cost remain open.

8 Conclusion

Quotient inputs buy exact protection over the class of fine graphs that share the quotient: worst-case regret over that class collapses to regret under the truth, while a single protected edge deletion already defeats fine-graph methods (Theorem 1). The price is the

action-space gap—zero under the support and confounding conditions of Cor. 1, linear in the horizon otherwise (Prop. 6), and recoverable by refinement only under the explicit conditions of Cor. 2. The same contract applies to arm-based, dynamic, and model-based CBO.

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A Proofs

Notation is as in §4.1 and §3.

Proof of Lemma 1. Write the run as decisions $d_1, d_2, \dots$ and datasets $\mathcal{D}_0, \mathcal{D}_1, \dots$, with

$$d_t = f_t(Q_\Pi(\mathcal{G}), \mathcal{D}_{t-1}, \omega_t, \eta).$$

When $Q_\Pi(\mathcal{G}) = Q_\Pi(\mathcal{G}')$, the first arguments agree. The fixed-objective protocol also fixes $\mathcal{D}_0$, the random stream, and the hyperparameters. Thus d_1 and the resulting $\mathcal{D}_1$ agree. Induction gives equality of every decision, dataset, and stopping event, hence of the full trajectories. Within-cluster edits and cross-cluster edits redundant in the quotient leave $Q_\Pi(\mathcal{G})$ unchanged, which proves the stated scope. $\square$

Lemma 2 (MIS reduction). *Let G be an ADMG with manipulable vertices M and target Y , let $\emptyset \neq \mathbf{W} \subseteq M$, and set $\mathbf{W}' = \mathbf{W} \cap \text{An}(Y)$ computed in $G_{\bar{\mathbf{W}}}$. Then (i) $P(Y \mid \text{do}(\mathbf{W} = \mathbf{w})) = P(Y \mid \text{do}(\mathbf{W}' = \mathbf{w}|_{\mathbf{W}'}))$ for every $\mathbf{w}$, (ii) $\mathbf{W}'$ is a MIS (or empty, when no member reaches Y), and (iii) consequently $\min_{\mathbf{X}_s \in \text{MIS}(G)} \mu^*(\mathbf{X}_s) = \min_{\emptyset \neq \mathbf{W} \subseteq M} \mu^*(\mathbf{W})$: restricting the search to MIS loses no value.*

Proof. (i) In $G_{\bar{\mathbf{W}}}$ the members of $\mathbf{W} \setminus \mathbf{W}'$ have no directed path to Y , so Y is unaffected by their values under the intervention. Dropping them from the intervention changes no mechanism on any directed path into Y (do-calculus rule 3). (ii) A directed path from $w' \in \mathbf{W}'$ to Y in $G_{\bar{\mathbf{W}}}$ passes through no intervened vertex (their incoming edges are cut), hence through no member of $\mathbf{W} \supseteq \mathbf{W}'$. The same path exists in $G_{\bar{\mathbf{W}'}}$, which only restores edges, so $\mathbf{W}' \subseteq \text{An}(Y)$ there. (iii) By (i) and (ii) every candidate’s value is attained by a MIS, and $\text{MIS}(G)$ is a subset of the candidates. $\square$

Proof of Prop. 1. Take $\mathbf{X} = \{a_1, a_2, b_1, b_2\}$ with fine edges $a_1 \rightarrow b_1$ and $b_2 \rightarrow a_2$. The coarsest partition $\Pi = \{\{a_1, a_2, b_1, b_2\}\}$ is valid: both edges are intra-cluster, so the quotient has a single manipulable node and is acyclic. Its refinement $\Pi' = \{\{a_1, a_2\}, \{b_1, b_2\}\}$ exposes both edges as inter-cluster ($a_1 \rightarrow b_1$ contributes $A \rightarrow B$ and $b_2 \rightarrow a_2$ contributes $B \rightarrow A$), inducing the 2-cycle $A \rightleftarrows B$, so $\mathcal{G}^{\Pi'}$ is cyclic and Π' is invalid. Decidability is immediate from the acyclicity clause of Def. 1. $\square$

Proof of Prop. 2. (i) Membership at Π' is the stated hypothesis (any union of Π -clusters is a union of Π' -clusters, so the candidate exists at Π' , and being a MIS of $\mathcal{G}^{\Pi'}$ puts it in $\mathbf{ES}^{\Pi'}$). For the identification functional, quotienting is compositional. The graph $\mathcal{G}^\Pi$ is the quotient of $\mathcal{G}^{\Pi'}$ under the induced partition of Π' -clusters, so every fine graph compatible with $\mathcal{G}^{\Pi'}$ is compatible with $\mathcal{G}^\Pi$. Identifiability from $\mathcal{G}^\Pi$ means a single functional of the observational distribution equals $\mathbb{E}[Y \mid \text{do}(\mathbf{X}_s)]$ across all SCMs whose graph is compatible with $\mathcal{G}^\Pi$ [Anand et al., 2023]. A fortiori the same functional works across the smaller class compatible with $\mathcal{G}^{\Pi'}$. MIS membership can fail after refinement because restoring distinctions between cluster members may make one member redundant under intervention. $\square$

Proof of Prop. 3. By Lemma 2(iii) applied to $\mathcal{G}^\Pi$, $V(\Pi) = \min_{\mathbf{X}_s \in \mathbf{ES}^\Pi} \mu^*(\mathbf{X}_s)$ equals the minimum of μ^* over all non-empty unions of Π -clusters (the value μ^* of a cluster union is a property of the underlying SCM, the same at every partition that can express the union). Unions of Π -clusters are unions of Π' -clusters and, at $\Pi_\perp$, all non-empty subsets of $\mathbf{X}$, so the three minima are over nested candidate families: $V(\Pi_\perp) \leq V(\Pi') \leq V(\Pi)$. Equality $V(\Pi) = V(\Pi_\perp)$ holds iff the unrestricted minimum is attained by a union of Π -clusters, and by Lemma 2(iii) at $\Pi_\perp$ the unrestricted minimum is min over Eq. (1)’s exploration set. This is exactly the stated condition. $\square$

Proof of Prop. 4. Fix A and a . Consistency and the assumed cross-world independence give

$$\begin{aligned} \mathbb{E}[Y_a] &= \int \mathbb{E}[Y_{a,b} \mid B_a = b] dP(B_a = b) \\ &= \int \mathbb{E}[Y_{a,b}] dP(B_a = b) \\ &\geq \inf_{b \in D_{B(a)}} \mathbb{E}[Y_{a,b}], \end{aligned}$$

where the support inclusion justifies the last line. Therefore

$$\mu^*(\text{cl}_\Pi(A)) \leq \mu^*(A).$$

The closure of every fine arm is a union of clusters, so $V(\Pi) \leq V(\Pi_\perp)$. Conversely, $\mathcal{U}(\Pi) \subseteq \mathcal{U}(\Pi_\perp)$ implies $V(\Pi) \geq V(\Pi_\perp)$. Hence equality holds. $\square$

Lemma 3 (Cluster containment implies cross-world independence). *In an NPSEM with mutually independent exogenous variables and augmented graph $\mathcal{G}^{\text{aug}}$: if Π has cluster-contained confounding (Def. 4), then for every $A \subseteq \mathbf{X}$, every $a \in D(A)$, and every $b \in D_B(a)$, $B_a \perp Y_{a,b}$.*

Proof. Under $\text{do}(A = a)$ the members of B retain their mechanisms, so recursive substitution in the mutilated model writes B_a as a measurable function of the exogenous ancestors of B in $\mathcal{G}_A^{\text{aug}}$ —that is, of $\mathbf{U}_B(A)$ (the clamped values a enter as constants). Under $\text{do}(\text{cl}_\Pi(A) = (a, b))$, recursive substitution writes $Y_{a,b}$ as a measurable function of $\mathbf{U}_Y(A)$, the exogenous ancestors of Y in $\mathcal{G}_{\text{cl}_\Pi(A)}^{\text{aug}}$. The two ancestor sets are disjoint by Def. 4, and the exogenous variables are mutually independent, so any function of the first is independent of any function of the second. $\square$

Proof of Cor. 1. Under Def. 3, the joint support of B_a lies in the product of its marginal supports, each contained in the corresponding $D(v)$, whose product is $D_B(a)$; hence $\text{supp}(B_a) \subseteq D_B(a)$. Lemma 3 gives $B_a \perp Y_{a,b}$, so Prop. 4 applies.

For the ADMG reading quoted after Def. 4: suppose some $U^* \in \mathbf{U}_B(A) \cap \mathbf{U}_Y(A)$. Then U^* has a directed path p_1 to a member of B avoiding the clamped A , and a directed path p_2 to Y avoiding the entire clamped closure. Let W be the closure member on p_1 nearest U^* ; the segment of p_1 from W back to U^* meets no other closure member. Splicing it with p_2 at U^* yields a backdoor path from the cluster member W to Y whose interior avoids every other member of W 's cluster—violating the backdoor phrasing. Hence the backdoor phrasing implies the exogenous-ancestor disjointness of Def. 4: checking it in the ADMG suffices.

Both premises can be necessary. In MEDIATEDCHAIN, setting $X_1 = 2$ lets the natural response of X_2 reach 4, while hard interventions constrain X_2 to $[-2, 2]$. The support condition fails and $\Delta(\Pi) = 3.96$. For an independence counterexample, let $C = \{A, B\}$, $B = U$, and $Y = -BU + (A - 1)^2$ with $U \sim \mathcal{N}(0, 1)$. Then $\mathbb{E}[Y \mid \text{do}(A = 1)] = -1$, whereas $\mathbb{E}[Y \mid \text{do}(A = a, B = b)] = (a - 1)^2 \geq 0$. Whole-cluster intervention severs the useful dependence, giving $\Delta(\Pi) = 1$ despite unrestricted domains. $\square$

Proof of Theorem 1. (i) Let $\mathcal{M}_\lambda$ be PARALLEL PARENT (App. B) with curvature $\lambda = \beta/4.25$: manipulable $\mathbf{X} = \{X_1, X_2\}$, edges $X_1 \rightarrow Y$ and $X_2 \rightarrow Y$, latent $X_1 \leftrightarrow X_2$, domains $[-3, 3]^2$, and $\Pi = \{\{X_1, X_2\}, \{Y\}\}$. Let H delete $X_1 \rightarrow Y$. The quotient is unchanged—the cluster edge $C_1 \rightarrow Y$ survives through $X_2 \rightarrow Y$ and $X_1 \leftrightarrow X_2$ is intra-cluster—so $H \in E_\Pi(\mathcal{G})$, and $\Delta(\Pi) = 0$ because $\mathcal{U}(\Pi) = \{\{X_1, X_2\}\}$ and the joint arm attains $y^* = 0$ at $(2, -2) \in [-3, 3]^2$. In H , however, X_1 has no directed path to Y , so $X_1 \notin \text{An}(Y)$ in every mutilation and Eq. (2) gives $\text{MIS}(H) = \{\{X_2\}\}$. By the closed form of App. B, $\mathbb{E}[Y \mid \text{do}(X_2 = x_2)] = 4.25\lambda + (x_2 + 2)^2 \geq 4.25\lambda = \beta$ for every x_2 . Every executed intervention—initial design included—is an $\{X_2\}$ -intervention, so every candidate incumbent satisfies $\mu(\hat{a}_t) \geq \beta$, i.e. $r_t \geq \beta$ and $R_T(\cdot; H) \geq \beta T$. The supremum over $E_\Pi(\mathcal{G})$ is at least this value. Both restrictions are necessary: an initial evaluation of the joint arm $\{X_1, X_2\} \notin \text{MIS}(H)$, or an incumbent chosen outside the executed set, could otherwise seed a zero-regret incumbent.

(ii) Trajectory identity across $E_\Pi(\mathcal{G})$ is Lemma 1: decisions and outcomes (drawn from the fixed $\mathcal{M}$) coincide step by step, so $R_T(\mathcal{A}; H) = R_T(\mathcal{A}; \mathcal{G})$ for every $H \in E_\Pi(\mathcal{G})$ and the supremum is attained everywhere. For the decomposition, add and subtract $V(\Pi)$ in each term of $R_T = \sum_t (\mu(\hat{a}_t) - y^*)$:

$$r_t = \underbrace{V(\Pi) - y^*}_{=\Delta(\Pi)} + (\mu(\hat{a}_t) - V(\Pi)).$$

The incumbent is an executed action on some $A \in \mathcal{U}(\Pi)$, and $V(\Pi)$ is the minimum of μ^* over $\mathcal{U}(\Pi)$ (Eq. (3)), so $\mu(\hat{a}_t) \geq V(\Pi)$; $\Delta(\Pi) \geq 0$ by Prop. 3. $\square$

Proof of Prop. 5. Constancy on $E_\Pi(\mathcal{G})$ for every $\mathcal{G}$ means $\tau(H)$ depends on H only through the fiber of Q_Π containing it. Setting $\bar{\tau}(q) := \tau(H)$ for any H with $Q_\Pi(H) = q$ is therefore well defined, and $\tau = \bar{\tau} \circ Q_\Pi$. $\square$

Proof of Prop. 6. By Lemma 2(iii), $y^* = V(\Pi_\perp)$. A fixed partition executes only arms in $\mathcal{U}(\Pi)$, so every incumbent satisfies

$$r_t = \mu(\hat{a}_t) - y^* \geq V(\Pi) - V(\Pi_\perp) = \Delta(\Pi).$$

Summation gives $R_T \geq T\Delta(\Pi)$. $\square$

Proof of Cor. 2. A visited Π_J with $\Delta(\Pi_J) = 0$ has an exploration set containing an optimal arm (Prop. 3 and Eq. (3)). Consistency in the sense of Def. 5 gives $\mu(\hat{a}_t) \rightarrow V(\Pi_J) = y^*$ almost surely, i.e. $r_t \rightarrow 0$, and Cesàro averaging gives $R_T/T \rightarrow 0$. $\square$

Proof of Cor. 3. Induct over phases. Within phase j , every graph-dependent component factors through $(\Pi_j, \mathcal{G}^{\Pi_j})$, while the trigger and split are functions of the trajectory and declared hierarchy. Equal phase- j quotients and prefixes therefore give the same phase trajectory by Lemma 1, followed by the same trigger and partition Π_{j+1} . Repeating through Π_J proves equality of the adaptive trajectories. An edit inside a never-split cluster remains invisible at every visited quotient. An edit inside a split cluster is invisible before the split and may become visible afterward. $\square$

Proof of Prop. 7. Singleton clusters make the quotient map the identity on $\mathcal{G}^*$, so $\mathcal{G}^\Pi = \mathcal{G}^*$, unions of singleton clusters are exactly the subsets of $\mathbf{X}$, and the MIS condition of Eq. (4) coincides verbatim with Eq. (2). The prior construction poses the same do-calculus queries CBO poses on the full graph, so priors coincide arm-for-arm. $\square$

Proof of Prop. 8. Under causal sufficiency the fine SCM is Markov relative to $\mathcal{G}$ with mutually independent exogenous noises, so for any intervention $\text{do}(\mathbf{x}_S)$ on a union of clusters S the fine truncated factorization gives $P(\mathbf{V} \mid \text{do}(\mathbf{x}_S)) = \prod_{k \notin S} P(X_k \mid \mathbf{X}_{\text{pa}_{\mathcal{G}}(k)}) \big|_{\mathbf{x}_S = \mathbf{x}_S}$. Group the factors by the clusters of Π . For a cluster $C \not\subseteq S$, the product of its members' factors is, by the chain rule applied in any topological order of $\mathcal{G}$ restricted to C , exactly the joint conditional $P(\mathbf{X}_C \mid \mathbf{X}_{\text{pa}(C)})$, where $\text{pa}(C)$, the union of C 's parent clusters, contains every out-of-cluster variable appearing in a member's conditioning set (validity of Π guarantees the quotient is acyclic, so $\text{pa}(C)$ is well defined and precedes C in C-DAG order). The fine factorization therefore equals the product over clusters of the joint cluster mechanisms with the clusters of S clamped at $\mathbf{x}_S$: the cluster-level truncated factorization. Note both uses of sufficiency: independent noises across clusters (no bidirected edges) and the absence of unobserved common causes that would make a member's conditional depend on variables outside $\text{pa}(C) \cup C$. Replacing $P(\mathbf{X}_C \mid \mathbf{X}_{\text{pa}(C)})$ by the product of its per-member marginals would be exact iff the members are conditionally independent given $\text{pa}(C)$, in particular when C contains no intra-cluster edges. This proves why the construction uses a joint mechanism. $\square$

Proof of Prop. 9. Take $\mathbf{X} = \{X_2, X_3\}$ as one root cluster C , target $Y = X_2 + \varepsilon_Y$ its child, all noises standard normal and independent (sufficiency holds, and any Π clustering $\{X_2, X_3\}$ is valid). Let M_1 have $X_2 = \varepsilon_2$ and $X_3 = aX_2 + \varepsilon_3$. Let M_2 have $X_3 = \varepsilon'_3$, $X_2 = a'X_3 + \varepsilon'_2$, with $a' = a/(1+a^2)$, $\text{Var}(\varepsilon'_3) = 1+a^2$, and $\text{Var}(\varepsilon'_2) = 1-a^2/(1+a^2)$. Both induce the same zero-mean Gaussian joint on (X_2, X_3) , with covariance $\begin{pmatrix} 1 & a \\ a & 1+a^2 \end{pmatrix}$, hence the same joint cluster mechanism (here, the same root-cluster distribution), and the same $P(Y \mid \text{do}(\mathbf{x}_C))$ for the whole-cluster intervention (both set X_2 directly). But under the sub-cluster intervention $\text{do}(X_3 = x)$: in M_1 , $X_2 = \varepsilon_2$ is unaffected, so $\mathbb{E}[Y \mid \text{do}(X_3=x)] = 0$ for all x . In M_2 , $X_2 = a'x$, so $\mathbb{E}[Y \mid \text{do}(X_3=x)] = a'x \neq 0$ for $x \neq 0$, $a \neq 0$. Every quotient-respecting input (the C-DAG, the joint cluster mechanisms, all whole-cluster interventional distributions) coincides between M_1 and M_2 , yet the sub-cluster query differs. No function of those inputs can answer it. $\square$

B MinimalBench: specification and oracles

The three SCMs share one parameterization: Gaussian roots, manipulable domains $[-3, 3]$ (except the MEDICATEDCHAIN policy box), per-variable unit intervention costs, target noise $\sigma_Y = 0.1$, and no clipping. Every arm value is closed-form. The structural module that defines them (`ccbo/minibench.py`) is the same one the methods, the perturbation registry, the structural tests, and Table 1's emitter consume, so specification and experiment cannot drift. Observational datasets (5,000 rows, seed 0, with every run using the first $n_{\text{obs}}=100$) are produced by released generators whose built-in oracle gates assert the Monte-Carlo arm values against the closed forms below (tolerance 0.05) at build time.

PARALLELPARENT (Experiment A). Latent variable U , manipulable variables X_1, X_2 , and target Y :

$$U = 0.4\epsilon_0, \quad X_i = U + 0.3\epsilon_i, \quad Y = \lambda(X_1 - 2)^2 + (X_2 + 2)^2 + \sigma_Y\epsilon_Y,$$

with $\lambda = 1$. The latent projection carries $X_1 \leftrightarrow X_2$ (corr = 0.64), and $\text{Var}X_i = 0.25$. Coarse partition $C_1 = \{X_1, X_2\}$. Closed forms: $\mathbb{E}[Y \mid \text{do}(x_1, x_2)] = \lambda(x_1 - 2)^2 + (x_2 + 2)^2$, so $y^* = 0$ at the joint arm. Also, $\mathbb{E}[Y \mid \text{do}(x_1)] = \lambda(x_1 - 2)^2 + 4.25$ (the free co-parent’s bowl plus its variance) and $\mathbb{E}[Y \mid \text{do}(x_2)] = 4.25\lambda + (x_2 + 2)^2$. Under A1 (delete $X_1 \rightarrow Y$) the fine MIS is $\{X_2\}$ alone. Since $X_1 \notin \text{An}(Y)$ in every mutilation, the best reachable fine value is $G(\lambda) = \lambda(4 + 0.25) = 4.25\lambda$, the analytic gap of §5.1. Condition A1 deletes $X_1 \rightarrow Y$ and is quotient-redundant. Condition A2 adds $X_1 \rightarrow X_2$, creating an intra-cluster bow with latent $X_1 \leftrightarrow X_2$. The $\text{do}(X_1)$ effect loses identification, but membership is intact. Condition A3 assumes $X_1 \leftrightarrow Y$ (quotient-visible: the C-DAG gains $C_1 \leftrightarrow Y$ and the cluster arm drops to the uninformative tier, and QCBO keeps running).

FRONTDOOR (Experiment B). Latent variable U , manipulable variables X_1, M , and target Y :

$$U = 0.4\epsilon_0, \quad X_1 = U + 0.3\epsilon_1, \quad M = 2X_1 + 0.2\epsilon_M, \\ Y = 2(1 - e^{-(M-1)^2/(2 \cdot 0.3^2)}) + U + \sigma_Y\epsilon_Y.$$

The latent projection carries $X_1 \leftrightarrow Y$. The target is a narrow well (width 0.3 over the 6-wide domain): prior corruption is a global-search phenomenon, and on a smooth quadratic the same corruption measurably costs nothing (§5.2). Fine MIS = $\{X_1\}, \{M\}$ (chain). The effect of $\text{do}(M)$ is identified by backdoor through X_1 , and $\text{do}(X_1)$ by the front-door functional through M . Gaussian smoothing gives the closed forms: $\mathbb{E}[Y \mid \text{do}(m)] = 2(1 - e^{-(m-1)^2/0.18})$, so $y^* = 0$ at $\text{do}(M=1)$. Moreover, $\mathbb{E}[Y \mid \text{do}(x_1)] = 2(1 - \frac{0.3}{\sqrt{0.3^2 + 0.2^2}} e^{-(2x_1-1)^2/(2(0.3^2+0.2^2))})$, best ≈ 0.336 at $x_1 = 1/2$. The observational mean is $\mathbb{E}[Y] \approx 1.637$. The coarse partition $\{X_1, M\}$ quotients to the bow $C_1 \rightarrow Y$, $C_1 \leftrightarrow Y$. The cluster arm runs on the uninformative tier in *every* condition, and setting M directly makes the partition lossless ($V(\Pi) = y^* = 0$). Condition B1 assumes a spurious intra-cluster confounder $X_1 \leftrightarrow M$: both fine arms lose identification (the backdoor set $\{X_1\}$ for $\text{do}(M)$ is invalidated by the induced collider path, the front-door premise for $\text{do}(X_1)$ by the confounded $X_1 \rightarrow M$ edge), while the quotient drops intra-cluster bidirected edges and is unchanged.

MEDIATEDCHAIN (Experiment C). There are no latent variables. The manipulable variables are X_1, X_2 and the target is Y :

$$X_1 = 0.5\epsilon_1, \quad X_2 = 2X_1 + 0.2\epsilon_2, \quad Y = (X_2 - 4)^2 + \sigma_Y\epsilon_Y,$$

with $D(X_1) = [-3, 3]$ and the *policy box* $D(X_2) = [-2, 2]$. Fine MIS = $\{X_1\}, \{X_2\}$. The joint is not minimal on a chain because mutilating X_2 severs X_1 from $\text{An}(Y)$ (Prop. 2), so the joint intervention exists only as the quotient’s cluster arm, which must clamp X_2 into the box. Closed forms: $\mathbb{E}[Y \mid \text{do}(x_1)] = (2x_1 - 4)^2 + 0.04$, so $y^* = 0.04$ at $\text{do}(X_1=2)$, which drives X_2 ’s natural response to 4, outside the box. We have $\mathbb{E}[Y \mid \text{do}(x_2)] = (x_2 - 4)^2$, whose best value over the box is 4. Hence $V(\Pi) = 4$ and $V(\Pi) - y^* = 3.96$ (Prop. 3). The support premise of Prop. 4 fails by construction. The declared hierarchy splits $C_1 = \{X_1, X_2\}$ into singletons. One split reaches the identity partition, whose MIS contains the optimal $\text{do}(X_1)$ arm.

Structural tests and oracle gates. Dataset-free structural tests assert, per condition, the exact fine MIS, the C-DAG signature (in)equalities, and the prior-tier flips claimed above, and end-to-end regressions assert QCBO’s trajectory invariance (arm choices and intervention values exactly, with incumbents matching to 10^{-8}) under every protected condition and its divergence under A3. The data generators refuse to write datasets whose Monte-Carlo arm values, correlations, or policy-box geometry contradict the closed forms.

C Fixed-objective protocol: per-method details

The invariant is that every method optimizes the *same* objective, the true SCM’s noise-free `intervention_function`, and only the *structure each method reasons with* is perturbed. We realize this per method as follows.

QCBO (finest and coarse). The objective is read from the original graph’s SEM, while the structure is supplied separately through the assumed graph name (`CoarsenedGraph(..., assumed_graph_name=variant)`).

Swapping the assumed graph changes the quotient, the exploration set, the identification formulas, and the prior, but never the SEM that generates outcomes. *Prior pipeline.* An identifiable arm’s prior mean instantiates the identified formula as a pipeline of GP-based plug-ins (backdoor $\rightarrow$ front-door $\rightarrow$ g -computation) on the bidirected-edge expansion of $\mathcal{G}^{\Pi}$. A non-identifiable arm, or an identifiable query the sound-but-incomplete pipeline cannot instantiate, shares the common uninformative prior, the observational mean of Y . Because clusters are manipulable-only and interventions assign full clusters, the cluster-level formulas expand exactly into fine-grained regressions on observational data.

CBO (full-DAG identification). CBO is QCBO’s backend at the identity partition (Prop. 7): it runs the same complete identification algorithm, only on the full DAG rather than a quotient. The injection is the same assumed-graph swap as for QCBO (the perturbed structure reaches the exploration set and the identification formulas that build the prior), while outcomes are always drawn from the true SEM. Its trajectory is identical to QCBO’s at the finest partition on every graph.

Exploration-set disclosure. Both methods use the MIS of the assumed graph (Eq. 2), computed by the released code and verified against hand-derived Lee–Bareinboim ground truth per benchmark (for the minimal suite, the per-condition sets are also asserted by the structural tests of App. B). The CBO authors’ shipped code deviates from that definition in two places, which we correct for both methods in the family suite: on ToyGraph it lists the non-minimal arm $\{X, Z\}$ (mutilating Z cuts $X \rightarrow Z$, so $\text{do}(X, Z) \equiv \text{do}(Z)$ and the joint arm is redundant), and on CoralGraph it truncates the list at subsets of size three (an ad-hoc cap; the definition yields all 31 non-empty subsets, as every manipulable variable keeps a directed path to Y through never-intervened non-manipulables under any mutilation).

HQCBO. Phase 1 is the QCBO configuration run by the same code path with the same seed (the trigger is a function of the coarse trajectory alone), so its pre-trigger prefix coincides exactly with the recorded QCBO trajectory, an equality asserted per seed. The refined phase re-enters the same backend with the split partition, the unioned exploration set, and the carried per-arm data (§3.2); the vendored loop forces an observation step at the start of a phase, a small bias *against* refinement that we keep.

Plateau trigger (exact rule). Let $b_0, b_1, \dots$ be the best-so-far incumbent values. The trigger fires at the first $t \geq k$ with $b_{t-k} - b_t < \delta$ (minimization; the mirrored difference for maximization), provided $0 < t < T$; the defaults are $k = 5$ and $\delta = 10^{-3}$ (PLATEAU_K, PLATEAU_DELTA in `ccbo/minimal_suite.py`, rule in `ccbo/benchmark.py`). On firing, the incumbent arm’s clusters are split along the declared hierarchy; a split whose quotient is cyclic is refused and the run stays coarse (Prop. 1). Each newly exposed arm is initialized with the protocol’s three interventional points.

Objective-identity verification. For all methods the decoupling holds by construction: outcomes are always evaluated on the true graph’s SEM, and the perturbation enters only through the separately-supplied assumed-graph variant (including assumed latents, as in A3 and B1). The released data generators additionally refuse to build any dataset whose Monte-Carlo arm values contradict the closed-form oracles (App. B): feeding a method a wrong graph changes only its causal reasoning, never the problem it is solving.

D Dataset graphs and their quotients

For every dataset in the paper, the figures below show the fine DAG with its coarse partition and the quotient C-DAG the method actually consumes, in the convention of Fig. 1: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters. They are regenerated from the released structure registry by `scripts/emit_dataset_tikz.py`, so the pictures cannot drift from the code. On PARALLEL PARENT, note how the intra-cluster latent $X_1 \leftrightarrow X_2$ disappears in the quotient while $C_1 \rightarrow Y$ survives any single parent-edge deletion. This is the redundancy behind Experiment A; the FRONTDOOR and ToyGraph pairs both quotient to the same cluster bow, illustrating Prop. 9.

E Implementation and reproducibility

All methods share one surrogate/acquisition backend (the vendored CBO loop), so comparisons differ only in the assumed structure. **Surrogate.** Each exploration-set arm carries a GPY GPRegression model with an RBF kernel (signal variance and lengthscale initialized to 1, ARD off) and observation noise fixed near zero (10^{-10} ,



Figure 5: **ParallelParent (MinimalBench, Exp. A)**. Fine DAG with coarse partition (left) and quotient C-DAG (right); drawing conventions as stated at the start of this appendix. The intra-cluster $X_1 \leftrightarrow X_2$ disappears in the quotient; deleting $X_1 \rightarrow Y$ leaves $C_1 \rightarrow Y$ alive through $X_2 \rightarrow Y$ (quotient-redundant).



Figure 6: **FrontDoor (MinimalBench, Exp. B)**. Fine DAG with coarse partition (left) and quotient C-DAG (right); drawing conventions as stated at the start of this appendix. The quotient is a bow ($C_1 \rightarrow Y$, $C_1 \leftrightarrow Y$): the cluster arm runs on the uninformative prior tier in every condition, so the assumed intra-cluster $X_1 \leftrightarrow M$ cannot change it.

matching the deterministic mechanisms). The causal prior supplies the do-calculus estimate as the GP *mean function*, so Δ -metrics isolate the effect of a wrong prior/exploration set from surrogate noise. **Acquisition.** Causal expected improvement [Garnett, 2023], maximized over the box-constrained intervention domain by gradient-based L-BFGS. **Protocol (minimal suite).** Unit per-variable intervention costs, 3 initial interventional points per arm, $T=50$ trials (60 on MEDIATEDCHAIN), 30 seeds, $n_{\text{obs}}=100$ from one shared dataset per SCM; the family suite follows each base paper’s own protocol (§5.4). Every run additionally logs the selected arm and intervention values per trial, so invariance is checked on *decisions*, not only on incumbent values. **Invariance tolerance.** Arm-choice and intervention-value sequences are compared exactly; incumbent trajectories at 10^{-8} (GP-solver floating point). On every protected condition of Table 1 the sequences are identical and $\sup |\Delta| = 0$; the quotient-visible control moves trajectories on every seed. **Artifacts.** The MinimalBench structural module (`ccbo/minibench.py`), its data generators with built-in oracle gates, the dataset-free structural tests and end-to-end invariance regressions, the suite runner (`scripts/run_minimal_suite.py`), the native two-phase HQCBO (`ccbo/minimal_suite.py`), and every table/figure emitter are released, so all reported numbers regenerate from the released SCMs and seeds.

Initialization-inclusive cost accounting. Reported cumulative costs exclude the initial design (three interventional points per arm on MinimalBench, ten on the CBO family; per-variable unit costs), so methods with different arm counts consume different unlogged budgets at equal T . Table 4 reconstructs, per method, the exploration-set size, initialization cost, logged cost, and total, and re-evaluates each method at the matched budget C^* —the smallest total across the group’s methods (HQCBO additionally pays three points per arm exposed at its per-seed split). The trajectory *through* initialization is unlogged and is not reconstructed: each curve starts at its post-initialization incumbent placed at its full initialization cost, and budgets below that are reported as unavailable. Two reversals are worth naming: on Synthetic, CBO’s final-value advantage over QCBO disappears at matched total cost, and on Coral, CBO’s 31-arm initialization (800 cost units) exceeds QCBO’s entire total budget. On FRONTDOOR the comparison moves the other way: fine CBO reaches the well within a smaller total budget. The DCBO and MCBO families follow their base papers’ trial-indexed protocols and are not re-costed.



Figure 7: **MediatedChain (MinimalBench, Exp. C)**. Fine DAG with coarse partition (left) and quotient C-DAG (right); drawing conventions as stated at the start of this appendix. The policy box restricts $D(X_2)$ to $[-2, 2]$; the fine optimum $\text{do}(X_1=2)$ drives X_2 to 4, outside the box.



Figure 8: **ToyGraph (CBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right); drawing conventions as stated at the start of this appendix. The quotient is a bow ($C_1 \rightarrow Y$, $C_1 \leftrightarrow Y$): $\text{do}(C_1)$ is not identifiable from the C-DAG, so the arm runs on the uninformative prior tier.

Reproducibility statement. The repository archives the complete raw per-seed traces behind every figure and table (`artifacts/traces/`: five compressed archives with per-file SHA256 checksums and a provenance manifest recording source commits, generating scripts, and settings), pip-freeze snapshots of both experiment environments with platform details and pinned external-stack commits (`envs/`), and a verification gate (`scripts/verify_traces.py`) that re-checks the archive’s file grid, recomputes every reported aggregate at displayed precision, and re-tests the exact invariance predictions. One command (`scripts/reproduce_paper.sh`) regenerates every figure and table from the archives into a fresh working directory and rebuilds this document.

F Full results

Tables 5 and 6 give the per-environment detail behind §5.4’s model-based rows: finals and wall-clock on MCBO’s own tasks, and the within-family invariance counts under the intra-cluster perturbation of the model’s graph view. Per-seed CSVs for every minimal-suite unit (trial, incumbent, cumulative cost, selected arm, and intervention values) regenerate from the archived traces (App. E).

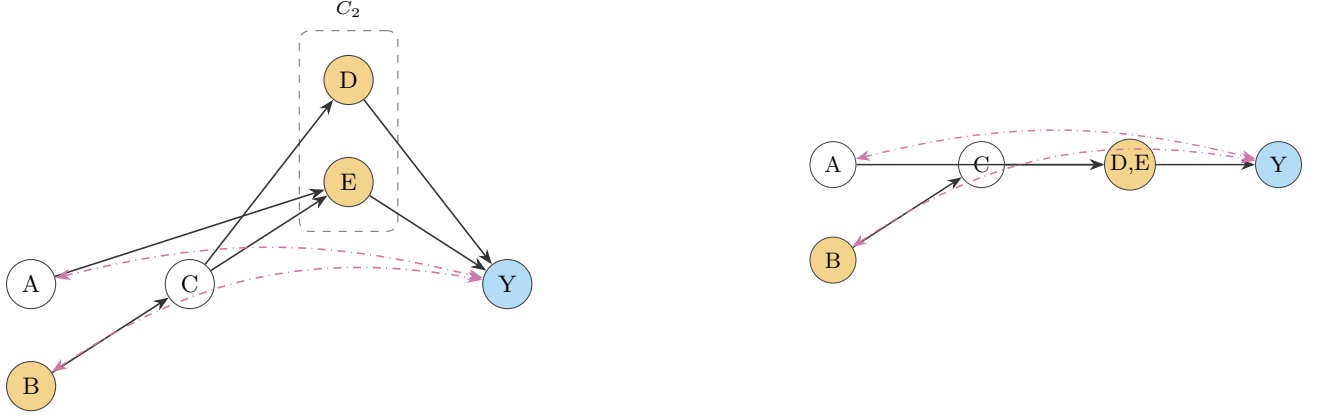


Figure 9: **CompleteGraph (CBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right); drawing conventions as stated at the start of this appendix.

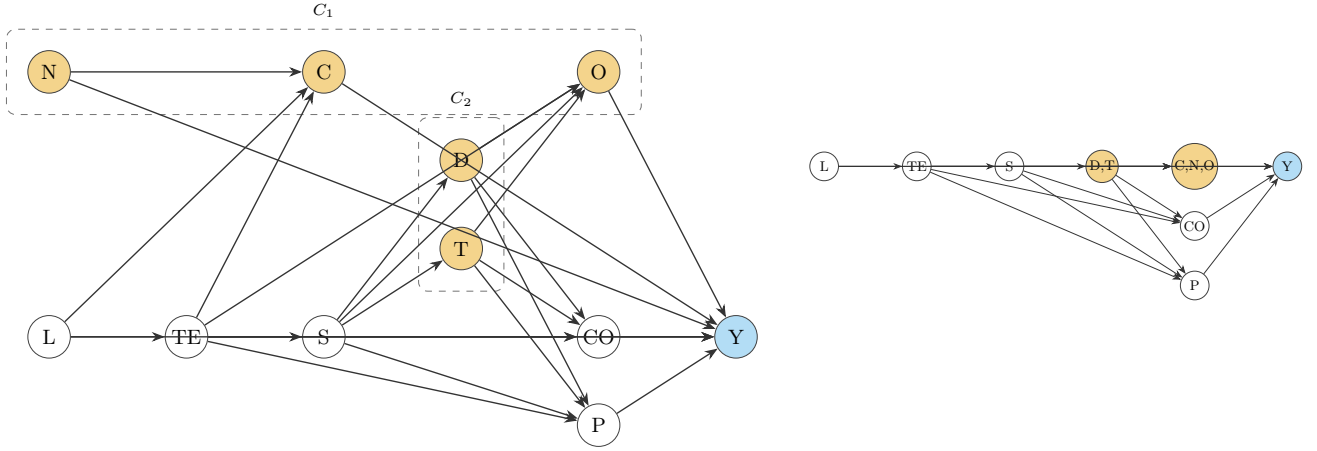


Figure 10: **SimplifiedCoralGraph (CBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right); drawing conventions as stated at the start of this appendix.

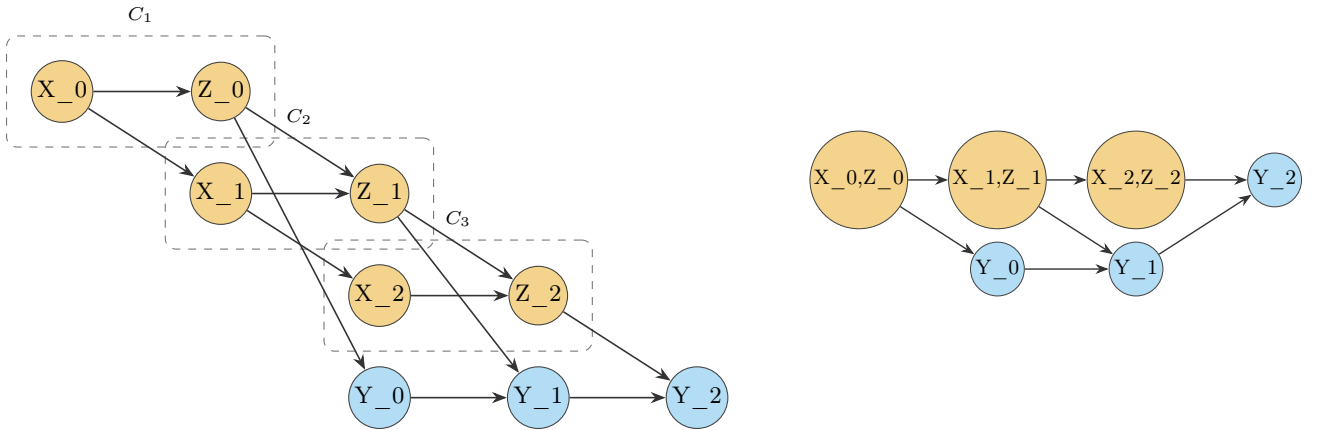


Figure 11: **DCBO stat / nonstat (QDCBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right); drawing conventions as stated at the start of this appendix. Slices $t = 0, 1, 2$; the nonstat setup shares this topology with a change point in the SEM.

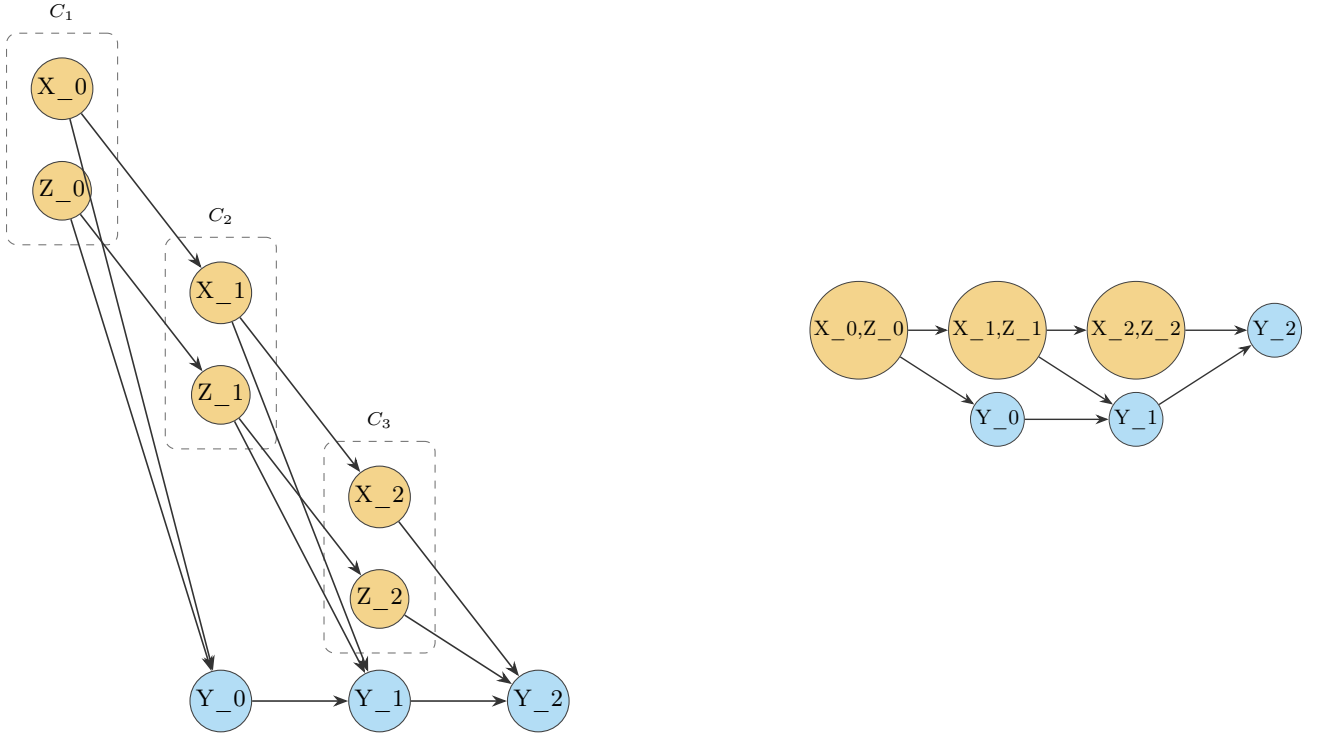


Figure 12: **DCBO ind (QDCBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right); drawing conventions as stated at the start of this appendix.



Figure 13: **ToyGraph-M (QMCBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right); drawing conventions as stated at the start of this appendix.

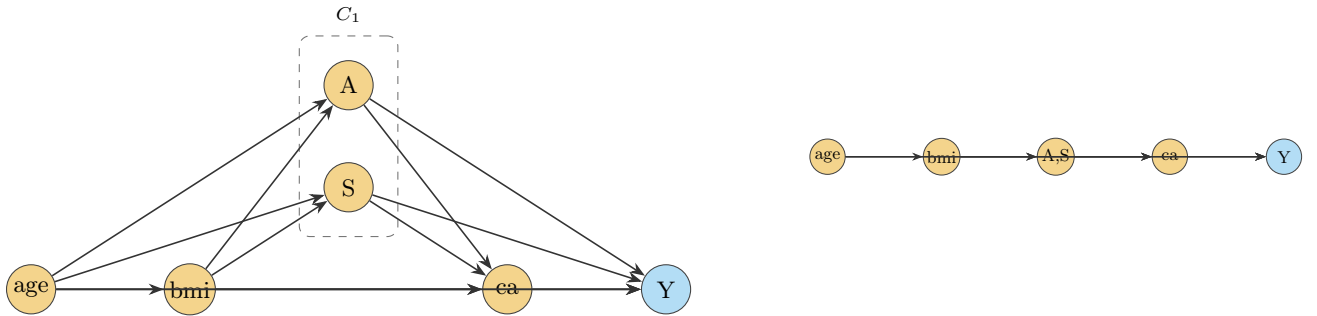


Figure 14: **PSAGraph (QMCBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right); drawing conventions as stated at the start of this appendix.

Algorithm 1 QCBO: Quotient Causal Bayesian Optimization

Require: C-DAG $\mathcal{G}^\Pi$ over the clusters of Π (or a fine hypothesis $\mathcal{G}$ and Π); target Y ; number of trials T ; observational data $\mathcal{D}^O$; optional hierarchy $\mathcal{H}$ (§3.2).

- 1: If given $\mathcal{G}$: $\mathcal{G}^* \leftarrow$ latent projection of $\mathcal{G}$; $\mathcal{G}^\Pi \leftarrow$ quotient of $\mathcal{G}^*$ under Π . Check acyclicity of $\mathcal{G}^\Pi$ (Assumption 1).
- 2: Build $\mathbf{ES}^\Pi \leftarrow \text{MIS}(\mathcal{G}^\Pi)$ via Eq. (4).
- 3: For each $\mathbf{X}_s \in \mathbf{ES}^\Pi$: run complete ID on $P(Y \mid \text{do}(\mathbf{X}_s))$ and fit the prior mean m_s via the two-tier rule, Eq. (5), on $\mathcal{D}^O$.
- 4: Collect initial interventional data $\mathcal{D}_0^I$.
- 5: **for** $t = 1, \dots, T$ **do**
- 6: Update GP posteriors $(m_{s,t}, k_{s,t})$ for all $\mathbf{X}_s \in \mathbf{ES}^\Pi$.
- 7: $(\mathbf{X}_{s_t}, \mathbf{x}_t) \leftarrow \arg \max_{s, \mathbf{x}} \text{EI}_t^s(\mathbf{x}) / \text{Co}(\mathbf{X}_s, \mathbf{x})$.
- 8: Intervene, observe y_t ; update $\mathcal{D}_t^I$.
- 9: **if** $\mathcal{H}$ declared **and** incumbent plateaued: split the incumbent’s clusters via $\mathcal{H}$, re-check validity, rebuild $\mathbf{ES}^\Pi \cup$ past arms, carry $\mathcal{D}_t^I$ by arm, init new arms (§3.2).
- 10: **end for**
- 11: **return** $(\mathbf{X}_{s_{t^*}}, \mathbf{x}_{t^*})$ where $t^* = \arg \min_t y_t$.

Table 4: Initialization-inclusive cost accounting and matched-budget comparison (means over seeds; Y at C^* by step interpolation above the post-initialization incumbent; “n/a” = initialization alone exceeds C^*).

Method	ES	Init	Logged	Total	Final Y	Y at C^*
<i>ParallelParent A0 (correct) ($C^*=73$)</i>						
CBO	3	12	83	95	0.003 ± 0.000	0.003 ± 0.001
QCBO	1	6	91	97	0.002 ± 0.000	0.003 ± 0.000
CEO	3	12	61	73	1.001 ± 0.304	1.011 ± 0.303
<i>ParallelParent A1 (del $X_1 \rightarrow Y$) ($C^*=49$)</i>						
CBO	1	3	46	49	4.245 ± 0.003	4.245 ± 0.003
QCBO	1	6	91	97	0.002 ± 0.000	0.004 ± 0.001
CEO	3	12	62	74	0.773 ± 0.258	0.794 ± 0.256
<i>FrontDoor B0 (correct) ($C^*=53$)</i>						
CBO	2	6	47	53	0.001 ± 0.000	0.001 ± 0.000
QCBO	1	6	94	100	0.001 ± 0.000	0.013 ± 0.004
CEO	3	12	58	70	0.070 ± 0.023	0.099 ± 0.030
<i>MediatedChain C0 ($C^*=61$)</i>						
CBO	2	6	55	61	0.040 ± 0.000	0.040 ± 0.000
QCBO	1	6	109	115	4.000 ± 0.000	4.000 ± 0.000
HQCBO	1	6+6	76	88	0.040 ± 0.000	0.040 ± 0.000
CEO	3	12	114	126	0.135 ± 0.035	0.213 ± 0.060
<i>Toy (family suite) ($C^*=52$)</i>						
CBO	2	20	32	52	-2.164 ± 0.000	-2.164 ± 0.000
QCBO	1	20	63	83	-2.131 ± 0.033	-2.071 ± 0.049
<i>Synthetic (family suite) ($C^*=96$)</i>						
CBO	6	90	62	152	-3.139 ± 0.190	-1.271 ± 0.280
QCBO	2	30	66	96	-1.245 ± 0.068	-1.245 ± 0.068
<i>Coral (family suite) ($C^*=217$)</i>						
CBO	31	800	94	894	36.078 ± 0.019	n/a
QCBO	3	100	117	217	36.417 ± 0.013	36.417 ± 0.013

Table 5: QMCBO vs MCBO (its own harness and protocol; 20 seeds, $T=100$). Final Y (all tasks maximized) and wall-clock per unit.

Env	Method	Final Y	s/unit
ToyGraph ($y^*=2.172$)	MCBO	1.392 ± 0.128	1270
	QMCBO	2.161 ± 0.004	1331
PSAGraph ($y^*=-4.13$)	MCBO	-5.153 ± 0.000	6789
	QMCBO	-5.153 ± 0.000	2455

Table 6: Invariance under an intra-cluster misspecification of the model’s graph view (same seeds, deterministic runs): trajectories identical to the correct-graph run, and mean $|\Delta_{\text{final}}|$. QMCBO cannot move (Lemma 1). MCBO does; on PSAGraph all twenty perturbed trajectories differ mid-run yet end within 0.001 of the same final.

Env	Method	identical	$ \Delta_{\text{final}} $
ToyGraph	MCBO	9/20	0.277
	QMCBO	20/20	0.000
PSAGraph	MCBO	0/20	0.001
	QMCBO	20/20	0.000

Table 7: Graph-uncertain comparator: CEO on the minimal suite. Shared with the fine methods: seeds 0–29, trial counts ($T=50/60$), three initial points per arm, the shared observational data, and the objective surface (noiseless targets are the closed-form $\mathbb{E}[Y \mid \text{do}(\cdot)]$, seam-checked per run against the true latent SEM). Not shared: total initialization cost (CEO initializes all subsets: 12 units vs. 6 for QCBO on PARALLEL PARENT), a different acquisition objective (with the same per-variable cost normalization), and learning from single stochastic true-SCM draws while the fine methods receive common-random-number Monte-Carlo mean evaluations (App. E gives initialization-inclusive totals). Pool: asserted observable DAG first plus every observable-distinct variant, uniform prior. A3/B1 add only an assumed latent confounder, so their observable edge sets equal A0’s/B0’s: those rows are aliases of the base runs. Δ is the per-seed paired difference (min task: positive = CEO worse); $P(\text{A0})$ is CEO’s mean final posterior mass on the baseline observable DAG: the true graph’s directed edge set with bidirected edges dropped (the true PARALLEL PARENT and FRONT DOOR graphs carry latent confounding that no candidate DAG can represent). Reported for multi-graph pools only.

Condition	CBO	QCBO	CEO	$\Delta(\text{CEO} - \text{QCBO})$ [95% CI]	$P(\text{A0})$
PP A0 (correct)	0.003 ± 0.000	0.002 ± 0.000	1.001 ± 0.304	+1.00 [+0.38, +1.62]	0.54
PP A1 (del $X_1 \rightarrow Y$)	4.245 ± 0.003	0.002 ± 0.000	0.773 ± 0.258	+0.77 [+0.24, +1.30]	0.55
PP A2 (add $X_1 \rightarrow X_2$)	0.002 ± 0.000	0.002 ± 0.000	0.712 ± 0.259	+0.71 [+0.18, +1.24]	0.54
PP A3 (assume $X_1 \leftrightarrow Y$) (= A0)	0.002 ± 0.000	0.001 ± 0.000	1.001 ± 0.304	+1.00 [+0.38, +1.62]	0.54
FD B0 (correct)	0.001 ± 0.000	0.001 ± 0.000	0.070 ± 0.023	+0.07 [+0.02, +0.12]	–
FD B1 (assume $X_1 \leftrightarrow M$) (= B0)	0.006 ± 0.002	0.001 ± 0.000	0.070 ± 0.023	+0.07 [+0.02, +0.12]	–
MC C0 (correct)	0.040 ± 0.000	4.000 ± 0.000	0.135 ± 0.035	–3.87 [–3.94, –3.79]	–